

300,000 acquisitions a day through the traditional paper-based process pan-India. But with our innovative eCAF, we are now getting almost 200,000 acquisitions in a day with just 10 per cent of our total retailers on-board,” Kumar reveals.

Likewise, the building of the Decision Tree app for optimising its call centre agents helped Airtel in reducing average call handling time. “The algorithm of the Decision Tree engine diagnoses the customer’s problem in real-time and provides insight to the agent on how to resolve it. The agent does not have to toggle between screens and multiple back-end applications, or check customer history and usage records to solve the query,” explains Kumar.

Decision Tree increased agent efficiency and reduced average handling time for calls by almost 50 per cent. These benefits also directly resulted in some unexpected cost savings for the telco.

Airtel has deployed MongoDB, Redis and Drools to enable Decision Tree for its agents.

Going forward, Airtel is also in the process of integrating its open source-backed eCAF solution within its payment bank. Kumar says that this move will help the masses open a bank account with Airtel instantly and support the ongoing digital payment movement in the country.

Community support to enhance developments

Airtel is using open source communities to improve its existing models and find ways to develop some innovations. “A great community can help to solve issues and provide a resolution to a problem,” Kumar states.

The engineering team members at Airtel’s premises mainly rely on Stack Overflow and GitHub. Kumar feels case studies are a vital source for gaining knowledge. “Case studies of companies using open source products provide us inputs on architecture and technology fitments,” he says.

Contributing back through strategic partnerships

In addition to leveraging community support, Airtel is contributing back to the open source world through some strategic partnerships with startups and enterprises that are building new open source based solutions. “We do not want to limit ourselves to the proprietary world at all, and would like to fully embrace open source; we are now even starting to contribute back,” Mehta proclaims.

California-based Aerospike is among the enterprises that has so far received support from Airtel. The startup offers a Flash-optimised in-memory open source NoSQL



Harmeen Mehta, global chief information officer, Bharti Airtel

database that operates in three distinct layers -- a Flash-optimised data layer, a self-managed distribution layer and a cluster-aware client layer.

Kumar told *Open Source For You* that his team used skills from Aerospike to address some issues in the telecom space. “Aerospike has open sourced its product for the developer community. We have involved the firm to help us crack complex problems in some of the core telco-domain applications,” he says.

The right mix of open source and proprietary software

Choosing an appropriate combination between open source and proprietary technologies is not child’s play. According to Kumar, the engineers at Airtel are using the following thumb rule. “Wherever we find that our application requirements are not being met by a licensed product, we go for open source,” states Kumar.

The team mainly contemplates open source solutions rather than proprietary technologies because of the former’s stability, available documentation and ease of implementation.

“We are really happy with the choices that we have made,” Mehta affirms. 

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